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A APPENDIX

A.1 PROOF FOR SATISFACTION OF MARGINAL CONSTRAINTS.

A.1.1 LAW OF LARGE NUMBERS

Proposition 1. *Given $(X_1, \dots, X_n)$, which are independently and identically distributed (IID) real d -dimension random variables, following a probability distribution $p(X)$, i.e., $X_i \sim p(X)$, $X \in \mathbb{R}^d$. We have an additional random variable Y that is random uniform sample of these variables, i.e., $P(Y = X_i) = \frac{1}{n}$. The cumulative distribution function (CDF) $\bar{F}(t)$ of random variable Y will converge to the $F(X)$, i.e., CDF of X .*

Proof: We first define an empirical cumulative distribution function $\hat{F}_n(X)$ over the random variables $(X_1, \dots, X_n)$:

$$\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{X_i \leq t}, \quad (4)$$

where $\mathbf{1}_{X_i \leq t}$ is an indicator for $X_i^d \leq t^d$ for all dimensions $\{1, \dots, d\}$.

The Glivenko–Cantelli theorem states that this empirical distribution function $\hat{F}_n(X)$ will converge to the cumulative distribution $F(X)$ if n is sufficiently large:

$$\sup_{t \in \mathbb{R}^d} |\hat{F}_n(t) - F(t)| \rightarrow 0. \quad (5)$$

If we have an additional random variable Y that its value is a random subsample of the variables $(X_1, \dots, X_n)$:

$$P(Y = X_i) = \frac{1}{n}, \forall i = 1, 2, \dots, n. \quad (6)$$

The CDF of this variable $\bar{F}(t)$ is:

$$\bar{F}(t) = P(Y \leq t) = \sum_{i=1}^n P(Y = X_i) \cdot \mathbf{1}_{X_i \leq t} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{X_i \leq t} = \hat{F}_n(t). \quad (7)$$

Therefore, the CDF of Y also converges to the original underlying CDF $F(t)$ if n is sufficiently large.

Proposition 2. *Assume we have n random samples $(X_1, \dots, X_n) \sim p_1$, and another n random samples $(Y_1, \dots, Y_n) \sim p_2$, and we are also given an arbitrary bijective map between random variables, i.e., $\Pi : \{1, \dots, n\} \leftrightarrow \{1, \dots, n\}$. If we construct a new random variable $Z : (X, Y)$ follows the following couplings:*

$$P(X = X_i, Y = Y_j) = \begin{cases} \frac{1}{n}, & \text{if } j = \Pi(i) \\ 0, & \text{else } j \neq \Pi(i); \end{cases}$$

The CDF of the marginal $P(X)$ will converge the CDF of p_1 , while the CDF of the marginal $P(Y)$ will converge to the CDF of p_2 .

Proof: Since Π is bijective, we can compute the marginal $P(X = X_i)$ directly:

$$\begin{aligned} P(X = X_i) &= \sum_{j=1}^n P(X = X_i, Y = Y_j) \\ &= P(X = X_i, Y = Y_{\Pi(i)}) + \sum_{j \neq \Pi(i)} P(X = X_i, Y = Y_j) \\ &= \frac{1}{n} + 0 = \frac{1}{n} \end{aligned} \quad (8)$$

Similarly, we can show the marginal of $P(Y)$ is also $\frac{1}{n}$. By leveraging Proposition 1, we show that $P(X)$ will converge the CDF of p_1 , and the CDF of $P(Y)$ will converge to the CDF of p_2 .

A.1.2 PROOF OF OUR OT APPROXIMATION

We first give a definition of coupling $q(x_0, x_1)$ in our case before showing its marginal fullfils the marginal requirements. In particular, we denote $x_0 \in R^{N \times 3}$ and $x_1 \in R^{N \times 3}$ as two random variables following the distributions, $q_0(x_0)$ and $q_1(x_1)$, respectively. It is noted that $q_0 := N(0, I)$, which is the standard Gaussian for each dimension in x_0 , and $q_1(x_1)$ is the distribution all possible point clouds, which involves the joint modeling of point cloud distribution given a shape S ($q_1(x_1|S)$) and the distribution of shape ($q(S)$), *i.e.*, $q_1(x_1) = \int q_1(x_1|S)q(S)dS$.

We can notice that each row in x_0 is independently and identically distributed (IID), *i.e.*, $q_0(x_0) = \prod_i^N \hat{q}_0(x_0^i)$, where we denote the i -th row of x_0 as x_0^i and distribution of x_0^i as $\hat{q}_0(x_0^i)$, which is 3-dimensional unit Gaussian. We can also assume each point in x_1 is IID given a shape, *i.e.*, $q_1(x_1|S) = \prod_i^N \hat{q}_1(x_1^i|S)$, where we denote the i -th row of x_1 as x_1^i and the distribution of x_1^i as $\hat{q}_1(x_1^i|S)$.

In our superset OT precomputation for a given shape S , we pre-sample a set of random variables $(x_0^1, \dots, x_0^j, \dots, x_0^M) \sim \hat{q}_0$, and a set of random variables $(x_1^1, \dots, x_1^k, \dots, x_1^M) \sim \hat{q}_1$, and have a precomputed bijective mapping $\Pi : \{1, \dots, M\} \leftrightarrow \{1, \dots, M\}$. With these defined, our coupling $\hat{q}(x_0^i, x_1^j|S)$ for one row in the training pair (x_0^i, x_1^j) given S can be formulated as:

$$\hat{q}(x_0^i = x_0^j, x_1^i = x_1^k|S) = \begin{cases} \frac{1}{n}, & \text{if } j = \Pi(k) \\ 0, & \text{else } j \neq \Pi(k); \end{cases}$$

Since the each row in the training pairs are independently subsampled, the coupling of the training pair (x_0, x_1) given a shape is defined as $q(x_0, x_1|S) = \prod_i^N \hat{q}(x_0^i, x_1^i|S)$. In the end, the coupling over all training pairs can be obtained by marginalize over all possible shapes, *i.e.*, $\int q(x_0, x_1|S)q(S)dS$.

Theorem 1. *Our coupling without blending converge the following marginal if the superset size M is sufficiently large:*

$$\int q(\mathbf{x}_0, \mathbf{x}_1)d\mathbf{x}_1 = q_0(\mathbf{x}_0), \int q(\mathbf{x}_0, \mathbf{x}_1)d\mathbf{x}_0 = q_1(\mathbf{x}_1). \quad (9)$$

Proof: We first show the left constraint:

$$LHS = \int q(x_0, x_1)dx_1 = \int \int q(x_0, x_1|S)q(S)dSdx_1 \quad (10)$$

$$= \int q(S) \int q(x_0, x_1|S)dx_1dS \quad \text{change the order of integration} \quad (11)$$

$$= \int q(S) \int \prod_i^N \hat{q}(x_0^i, x_1^i|S)d(x_1^1, \dots, x_1^N)dS \quad \text{independent assumption of each row in training pair} \quad (12)$$

$$= \int q(S) \prod_i^N \int \hat{q}(x_0^i, x_1^i|S)dx_1^i dS \quad \text{integrals of independent products} \quad (13)$$

$$= \int q(S) \prod_i^N \sum_k^M \hat{q}(x_0^i, x_1^k|S)dS \quad \text{restricting to discrete values in supersets} \quad (14)$$

$$= \int q(S) \prod_i^N \hat{q}_0(x_0^i)dS \quad \text{Proposition 2} \quad (15)$$

$$= \int q(S)q_0(x_0)dS = q_0(x_0) \quad \text{independent assumption of each row in Gaussian noises} \quad (16)$$

$$(17)$$